OSIPI · GSOC 2026 · ASL QUALITY CONTROL

Kidney QC the short version

Extending the ASL QC ToolBox from brain to kidney

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KIDNEY_QC_ONE_PAGER.md · 1,339 words

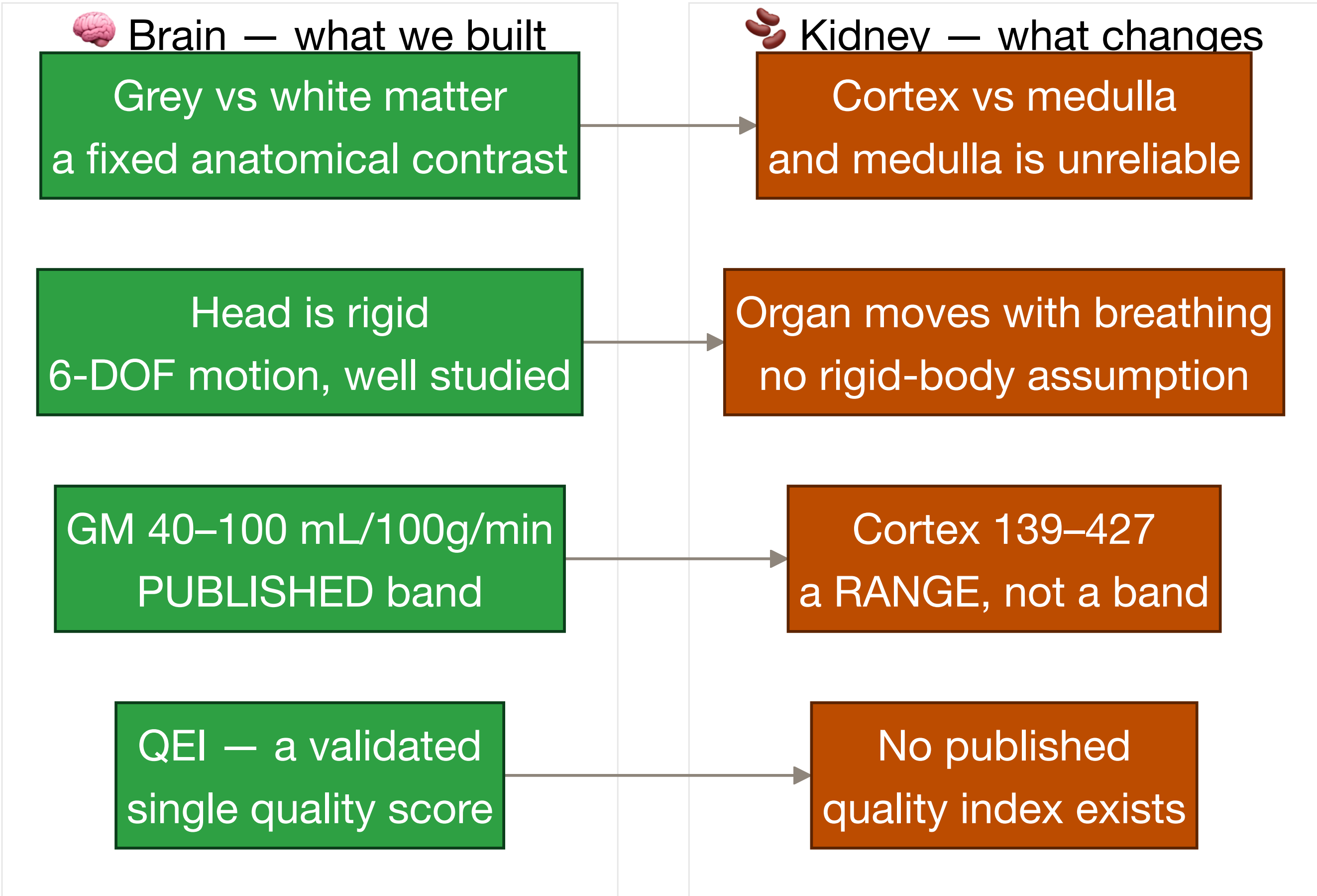
reconciled against osipy-qc · 20 registered checks · 253 tests

🔭 In one paragraph

The brain toolbox grades a CBF map against grey matter. The kidney has no grey matter. It has a **cortex and a medulla**, it **moves with every breath**, and it is perfused roughly **five times harder** than the brain. Most of the maths survives that move; almost none of the *thresholds* do. The single most important finding from reading the literature is a negative one, and it shapes everything below.

> ### 🚨 The finding that shapes the design
>
> **Nery et al. 2020** (*MAGMA*, PARENCHIMA COST Action) is the renal equivalent of the ASL White Paper — 59 consensus statements, and I read every one. It contains **zero numeric quality thresholds.** No tSNR cutoff, no CoV cutoff, no motion limit, no QEI equivalent.
>
> So the published material is all about *how to acquire* the scan, and none of it is about *whether the map is good*. v1 grades the map and leaves protocol conformance out of scope, which means every threshold we ship is labelled **UNCALIBRATED**. That is a publishable statement about the field, not a gap in our work.

🧠 → 🫘 Why the kidney is harder



The perfusion numbers do not converge. Published healthy cortical perfusion spans **139–427 mL/100 g/min**. The dominant driver is not biology — it is the **labelling scheme**: **FAIR reads 1.5–1.8× higher than pCASL in the same subjects** (362 ± 57 vs 201 ± 72). Field strength adds ~10%, age another ~15–20%. A band that is not conditioned on labelling scheme and field strength is measuring the protocol, not the patient.

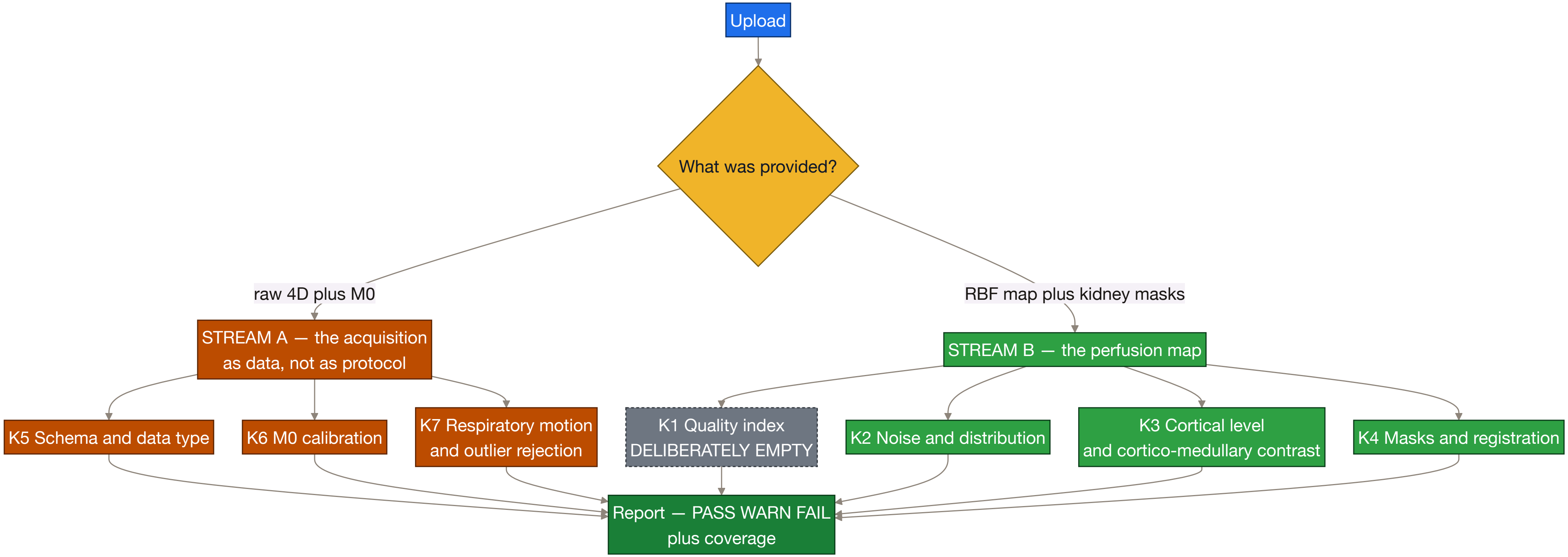
⚖️ Two consensus rules we get for free

These are the only *published mandates* that map directly onto our report, and both are cheap to honour:

RULE	AGREEMENT	WHAT IT MEANS FOR US
R10.1 — report **cortical** RBF (not whole-kidney), **separately for left and right**	**100%**	our report schema is decided for us: per-kidney, cortex-restricted
R10.2 — medullary values **"are not considered reliable with current measurement approaches"**	**89%**	🚨 the cortex:medulla ratio **cannot be a perfusion verdict**

R10.2 is the one that hurts. The cortex:medulla ratio looked like the obvious kidney analogue of the brain's GM/WM ratio — scale-free, contrast-based, exactly the kind of check that survives a calibration error. Consensus says the medullary half of it is not trustworthy. So it ships as a **segmentation-integrity flag** (INFO/WARN, never FAIL): if the ratio inverts, the *masks* are probably swapped — which is a real and useful thing to catch, just not the thing we hoped for.

🗺️ The proposed check set


Upload
What was provided?
raw 4D plus M0
RBF map plus kidney masks
STREAM A — the acquisition as data, not as protocol
STREAM B — the perfusion map
K5 Schema and data type
K6 M0 calibration
K7 Respiratory motion and outlier rejection
K1 Quality index DELIBERATELY EMPTY
K2 Noise and distribution
K3 Cortical level and cortico-medullary contrast
K4 Masks and registration
Report — PASS WARN FAIL plus coverage

Protocol conformance is deliberately out of scope for v1. Nery's 59 statements are the only published, directly checkable requirements in renal ASL — labelling scheme, timings, geometry, readout, quantification constants — and v1 does not grade any of them. That is a real cost, stated plainly: it leaves the design resting almost entirely on **uncalibrated** thresholds. The boundary is the same one the brain tool draws — grade the data, not the acquisition parameters.

K1 is deliberately empty for a different reason: there is no renal QEI, and inventing one would be the single least defensible thing we could do.

K7 carries the one genuinely implementable published rule family — subtraction-outlier rejection. Respiratory motion is the dominant renal artefact, and unlike the brain there is no rigid-body assumption to lean on.

📥 What you must bring

The kidney has no equivalent of "just run BET on it" — **a perfusion map alone buys almost nothing**, because every meaningful metric is ROI-restricted and the ROIs must come from outside.

TIER	SUPPLY	UNLOCKS
B0	RBF map only	almost nothing — histogram shape, implausible-value fraction
B1	+ whole-kidney masks, **left and right separately**	negative fraction, left-vs-right consistency, mask integrity
B2	+ **cortex mask per kidney**	**cortical RBF — the quantity R10.1 mandates**
B3	+ medulla mask (or a T1 map)	cortex:medulla ratio, as an integrity flag only
A0	raw 4D + kidney masks	outlier rate, respiratory displacement, control/label order
A1	+ M0	M0 checks, perfusion signal as % of M0
A2	+ acquisition metadata	four existing checks sharpen; **no new check** — v1 grades no protocol parameters
A3	+ respiratory trace	gating efficiency, motion-vs-outlier correlation

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🔧 What transfers from the brain code

BRAIN CHECK	VERDICT
Negative-voxel fraction, histogram	✅ **transfers as-is** — non-physical is non-physical in any organ
Schema, data type, M0 presence / TR / background suppression	✅ **transfers**, with renal metadata fields

BRAIN CHECK	VERDICT
CBF level, left-right consistency	🔶 **new bands** — and they must be conditioned on labelling scheme
GM/WM ratio → cortex:medulla	🔶 **reworked** — demoted to an integrity flag by R10.2
Motion (FWD/DVARS)	🔶 **reworked** — the organ moves, the head does not
QEI	❌ **does not transfer** — it is built on GM/WM/CSF probability maps and a grey-matter spatial prior. No renal equivalent exists

The architecture change this forces: `cfg.organ` already exists in the config but **no check reads it** — it is inert today. Kidney is the reason to make it real, so checks, thresholds and profiles become organ-aware rather than kidney being bolted on beside brain.

❓ For the mentors and the renal specialist

1. **R10.2 says medullary RBF is unreliable — do you agree the cortex:medulla ratio should be an integrity flag rather than a quality verdict?** It is the most useful-looking metric we have and consensus undercuts it.
2. **Should cortical bands be conditioned on labelling scheme?** FAIR vs pCASL differ 1.5–1.8× in the same subjects. One band cannot serve both, and pretending otherwise grades the protocol.
3. **Is there an unpublished or in-progress renal quality index we should build toward** rather than leaving K1 empty?
4. **What is realistically available in practice** — do sites have cortex and medulla masks, or is B1 (whole-kidney only) the honest default tier?

Full detail: `BRAIN_TO_KIDNEY.md` → `KIDNEY_ASL_EXPLAINED.md` → `KIDNEY_QC_DESIGN.md` **Read first:** `papers/nery2020_renal_asl_consensus.pdf` 51 papers · 68 highlighted source pages